The Bennett mode and its relation to nonlocal linear response

These are notes on the following paper by **Alan Bennett** (1970): Influence of the Electron Charge Distribution on Surface-Plasmon Dispersion (https://journals.aps.org/prb/pdf/10.1103/PhysRevB.1.203). In all calculations below, we use the notation of the **Bennett** paper. In **Section 1** below, we derive **Equation 2.12**, the central result, of this paper. In **Section 2**, we derive **Equation 2.14** of the paper. This equation gives the additional boundary condition (ABC) necessary to solve the modes of the system. In **Section 3**, we derive **Equation 3.1** and **Equation 3.2**, which give the surface plasmon mode profiles deep in the metal region (away from the metal-air interface). In **Section 4**, we show how overall charge neutrality imposes a restriction on the behavior of the ionic/electronic charge densities. Finally, in **Section 5**, we derive the electric field in steady state (**Equation 2.15** of the **Bennett** paper).

Section 1: Derivation of Equation 2.12

In this section we derive **Equation 2.12**. We start with **Equation 2.6** (the charge conservation equation) in conjunction with **Equation 2.7** (the force equation), which gives us the following differential equation:

$$i\omega n_1(\mathbf{r}) = \frac{n_0}{i\omega m n_0}\nabla\cdot\Bigg[en_1(\mathbf{r})E_0(\mathbf{r}) + en_0 f(\mathbf{r})E_1(\mathbf{r}) + \nabla p_1(\mathbf{r})\Bigg],$$

where $n_1(\mathbf{r})$ is the first order change in the electronic density, $E_0(\mathbf{r}) \equiv E_0(z)\mathbf{e}_z$ is the electric field in steady state, $n_0 f(\mathbf{r}) \equiv n_0 f(z)$ is the electronic density in steady state, $E_1(\mathbf{r}) = -\nabla\phi(z)e^{ik_x+ik_y}$ is the first order change in the electric field and $p_1(\mathbf{r}) = m\beta^2 n_1(\mathbf{r})$ is the first order change in the pressure of the system. Note that the electric field and electronic density in steady state are only functions of z. The latter claim is taken as an ansatz (see **Figure 2** of the paper) and the former claim follows from the former (see **Section 4** and **Section 5** below). We use **Equation 2.8** (Gauss's law) as well as the relation between the first order change in pressure to the first order change in electronic density to rewrite the equation above as follows:

$$\frac{\varepsilon_0}{e}\omega^2\nabla\cdot E_1(\mathbf{r}) = \frac{1}{m}\nabla\cdot\left[-\varepsilon_0 E_0(\mathbf{r})\nabla\cdot E_1(\mathbf{r}) + en_0 f(\mathbf{r})E_1(\mathbf{r}) - \frac{\varepsilon_0 m\beta^2}{e}\nabla\nabla\cdot E_1(\mathbf{r})\right]$$

We now write this equation in terms of the potential (see **Equation 2.10** of the **Bennett** paper), which gives us the following:

$$-\frac{\omega^2}{\omega_p^2}\left[\phi''(z) - k^2\phi\right] = \nabla\cdot\left[\frac{\varepsilon_0}{en_0}E_0(z)\left[\phi''(z) - k^2\phi(z)\right]e^{ik_x x + ik_y y}\mathbf{e}_z + \right.$$
$$\left. - f(z)\nabla\left[\phi(z)e^{ik_x x + ik_y y}\right] + \frac{\beta^2}{\omega_p^2}\nabla\left[\left[\phi''(z) - k^2\phi(z)\right]e^{ik_x x + ik_y y}\right]\right],$$

where we also multiplied both sides of the equation by m/en_0 and introduced the plasma frequency, ω_p, which is given by $\omega_p^2 = n_0 e^2/m\varepsilon_0$. Simplifying the equation above yields the following differential equation:

$$\frac{\beta^2}{\omega_p^2}\left[\left[\phi''''(z) - k^2\phi''(z)\right] - k^2\left[\phi''(z) - k^2\phi(z)\right]\right] - f(z)\left[\phi''(z) - k^2\phi(z)\right] - f'(z)\phi'(z) +$$
$$\frac{\varepsilon_0 E_0(z)}{en_0}\left[\phi'''(z) - k^2\phi'(z)\right] + \frac{\varepsilon_0}{en_0}E_0'(z)\left[\phi''(z) - k^2\phi(z)\right] = -\frac{\omega^2}{\omega_p^2}\left[\phi''(z) - k^2\phi(z)\right]$$

We rewrite this in terms of the quantity, $\varepsilon(z)$, which is defined in the **Bennett** paper. Using $\varepsilon(z) \equiv eE(z)/(m\omega_p^2) = eE(z)/(mn_0e^2/(m\varepsilon_0)) = \varepsilon_0 E(z)/(en_0)$ the equation above may be re-expressed as follows:

$$\phi''''(z) = \left[2k^2 - \frac{\omega^2}{\beta^2} + \frac{\omega_p^2}{\beta^2}f(z) - \frac{\omega_p^2}{\beta^2}\varepsilon'(z)\right]\phi''(z) +$$
$$\left[\frac{k^2\omega^2}{\beta^2} - k^4 - f(z)\frac{\omega_p^2 k^2}{\beta^2} + \frac{k^2\omega_p^2}{\beta^2}\varepsilon'(z)\right]\phi(z) + \left[\frac{\omega_p^2}{\beta^2}f'(z) + \frac{k^2\omega_p^2}{\beta^2}\varepsilon(z)\right]\phi'(z) - \frac{\omega_p^2\varepsilon(z)\phi'''(z)}{\beta^2},$$

which is **Equation 2.12** of the **Bennett** paper.

Section 2: Derivation of Equation 2.14

Equation 2.14 gives the additional boundary condition (ABC) necessary for finding the modes when longitudinal oscillations in the metal are allowed. This additional boundary condition requires $\mathbf{v}\cdot\mathbf{e}_z\Big|_{z=0^-} = 0$. In this case, **Equation 2.7** gives us the following:

$$en_1(z=0^-)E_0(z=0^-) + en_0 f(z=0^-)E_1(x,y,z=0^-) + \partial_z p(x,y,z=0^-) = 0$$

Due to overall charge neutrality, we have that $E_0(z=0^-)=0$. Furthermore, from **Figure 2**, we see that $f(z=0^-)=0$. Thus we see that the boundary condition imposed by **Equation 2.7** and the vanishing of the normal component of the current at the metal-air interface gives us the following:

$$\partial_z p(x,y,z=0^-) = 0 \rightarrow \partial_z \left[\nabla^2 \phi(\mathbf{r})\right] = 0 \rightarrow \partial_z \left[\phi''(z) - k^2\phi(z)\right] = 0 \rightarrow$$

$$\phi'''(z) - k^2\phi'(z) = 0,$$

where the last relation is **Equation 2.14** of the **Bennett** paper.

Section 3: Derivation of Equation 3.1 and Equation 3.2

Equation 3.1 and **Equation 3.2** give the spatial dependencies of the modes for $z < -a$, the region in which the total charge density is constant. For $f(z)=1$ (constant charge density), $\varepsilon(z)=0$, and $\varepsilon'(z)=0$ (constant null electric field which follows from overall charge neutrality of the system), we obtain the following differential equation for the electric potential (using **Equation 2.12** of the **Bennett** paper):

$$\phi''''(z) = \left[2k^2 + \frac{(\omega_p^2-\omega^2)}{\beta^2}\right]\phi''(z) - \left[k^4 + k^2\frac{(\omega_p^2-\omega^2)}{\beta^2}\right]\phi(z)$$

To find the allowed modes, we assume an exponential dependence: $\phi(z) \propto e^{qz}$, which gives us the following fourth order equation for the wavenumber, q:

$$q^4 - \left[2k^2 + \frac{(\omega_p^2-\omega^2)}{\beta^2}\right]q^2 + \left[k^4 + k^2\frac{(\omega_p^2-\omega^2)}{\beta^2}\right] = 0 \rightarrow$$

$$q^2 = \frac{2k^2 + (\omega_p^2-\omega^2)/\beta^2 \pm \sqrt{4k^4 + 4\frac{k^2(\omega_p^2-\omega^2)}{\beta^2} + \frac{(\omega_p^2-\omega^2)^2}{\beta^4} - 4k^4 - 4\frac{k^2(\omega_p^2-\omega^2)}{\beta^2}}}{2}$$

We thus obtain two possible values for q^2 (and thus four possible values for q) given as follows:

$$q^2 = \frac{2k^2 + (\omega_p^2-\omega^2)/\beta^2 \pm (\omega_p^2-\omega^2)/\beta^2}{2} = k^2, (\beta^2 k^2 + \omega_p^2 - \omega^2)/\beta^2$$

These are the solutions given by **Equation 3.1** and **Equation 3.2**, as desired.

Section 4: Restrictions imposed by the overall charge neutrality of the system

The charge density of the electrons, $n(\text{electrons})$, and the charge density of the ions, $n(\text{ions})$, are given by the following expressions:

$$n(\text{electrons}) = -en_0\left[\Theta(-a-z) - \Theta(-z)\Theta(z+a)\frac{z}{a}\right]$$
$$n(\text{ions}) = en_0\left[\Theta(-b-z) - \Theta(b+z)\Theta(-c-z)\left[\frac{c+z}{b-c}\right]\right],$$

where Θ is the Heaviside step function. In the expression for the electronic charge density, the first term ensures the charge density is equal to $-en_0$ for $a+z<0 \rightarrow z<-a$. Furthermore, the second term ensures that the charge density goes as $en_0 z/a$ in the region $-a<z<0$, consistent with **Figure 2** of the **Bennett** paper. In the expression for the ionic charge density, the first term ensures that the charge density equals en_0 for $z<-b$. Furthermore, the second term ensures that the charge density equals $(-en_0)(c+z)/(b-c)$ in the region $-b<z<-c$.

The total charge density (which must be zero for a charge neutral system) will be given by (ignoring the obviously charge neutral region for $z<-a$):

$$en_0\left[(-a/2)+(a-b)+(b-c)/2\right] = 0 \rightarrow a/2 - b/2 - c/2 = 0 \rightarrow a-c=b,$$

where the last equality is the only restriction that charge neutrality imposes on the electronic and ionic charge densities.

Section 5: The electric field in steady state (Equation 2.15 of the Bennett paper)

In this section, we calculate the electric field in steady state (denoted by $E_0(z)$ in **Section 1**) in the different regions of the metal-air system.

In the region $-a<z<-b$, the electric field from the positively charged ions (which we denote by $E(\text{ions})$) is given by the following expression:

$$E(\text{ions}) = \frac{en_0}{2\varepsilon_0}\left[(b+z)+(z+a)-\frac{(b-c)}{2}\right],$$

where $(z+a)(en_0)$ is the effective surface charge density to the left of z and $(-b-z)en_0+(-c+b)en_0/2$ is the effective surface charge density to the right of z. The electric field from the negatively charged electrons (which we denote by $E(\text{electrons})$ will be given by a similar expression:

$$E(\text{electrons}) = -\frac{en_0}{2\varepsilon_0}\left[-\frac{z^2}{2a}+\left[\frac{a}{2}-\frac{z^2}{2a}\right]\right],$$

where $(-en_0)(-z)(-z/2a)$ is the effective surface charge to the right of z and $(-en_0)(a/2-z^2/2a)$ is the effective surface charge to the left of z. Thus the total field (engendered by both the positively charged ions and the negatively charged electrons) will be given by the following expression:

$$E(\text{total}) \equiv E(\text{ions})+E(\text{electrons}) = \frac{en_0}{2\varepsilon_0}\left[2z+\frac{a+b+c}{2}+\frac{z^2}{a}\right] =$$

$$\frac{en_0}{2\varepsilon_0}\left[2z+\frac{z^2}{a}+a\right] = \frac{en_0}{2\varepsilon_0 a}(z+a)^2,$$

where we used $a-c=b$ (which is an identity which we derived in **Section 4**) in the second to last equality. We may re-express the above as follows:

$$\varepsilon \equiv \frac{eE(\text{total})}{m\omega_p^2} = \frac{e\varepsilon_0 E(\text{total})}{n_0 e^2} = \frac{(z+a)^2}{2a},$$

which is the second line of **Equation 2.15**.

In the region $-b<z<-c$, we have, similarly,

$$E(\text{ions}) = \frac{en_0}{2\varepsilon_0}\left[(a-b)+\left[\frac{b-c}{2}-\frac{(c+z)^2}{2(b-c)}\right]-\frac{(c+z)^2}{2(b-c)}\right] \to$$

$$E(\text{electrons})+E(\text{ions}) = \frac{en_0}{2\varepsilon_0}\left[-\frac{(c+z)^2}{(a-2c)}+\frac{z^2}{a}\right] = \frac{en_0}{2\varepsilon_0(a-2c)}\left[-c^2-2cz-2cz^2/a\right]$$

$$= -\frac{en_0}{\varepsilon_0(a-2c)}\left[\frac{c^2}{2}+cz+\frac{cz^2}{a}\right],$$

which is the third line of **Equation 2.15**.

In the region $-c<z$, we have, similarly,

$$E(\text{ions}) = \frac{en_0}{2\varepsilon_0}\left[a-b/2-c/2\right] = \frac{en_0}{2\varepsilon_0}\frac{a}{2}, E(\text{electrons}) = -\frac{en_0}{2\varepsilon_0}\left[\frac{a}{2}-\frac{z^2}{a}\right] \to$$

$$E(\text{total}) = \frac{en_0}{2\varepsilon_0}\frac{z^2}{a},$$

which is the last line of **Equation 2.15**.